Ecommerce Recommendation System

Welcome to our presentation on the cutting-edge Ecommerce Recommendation System, designed to revolutionize online shopping. Join us to discover the future of personalized product recommendations.



Existing System

1 Generic Recommendations

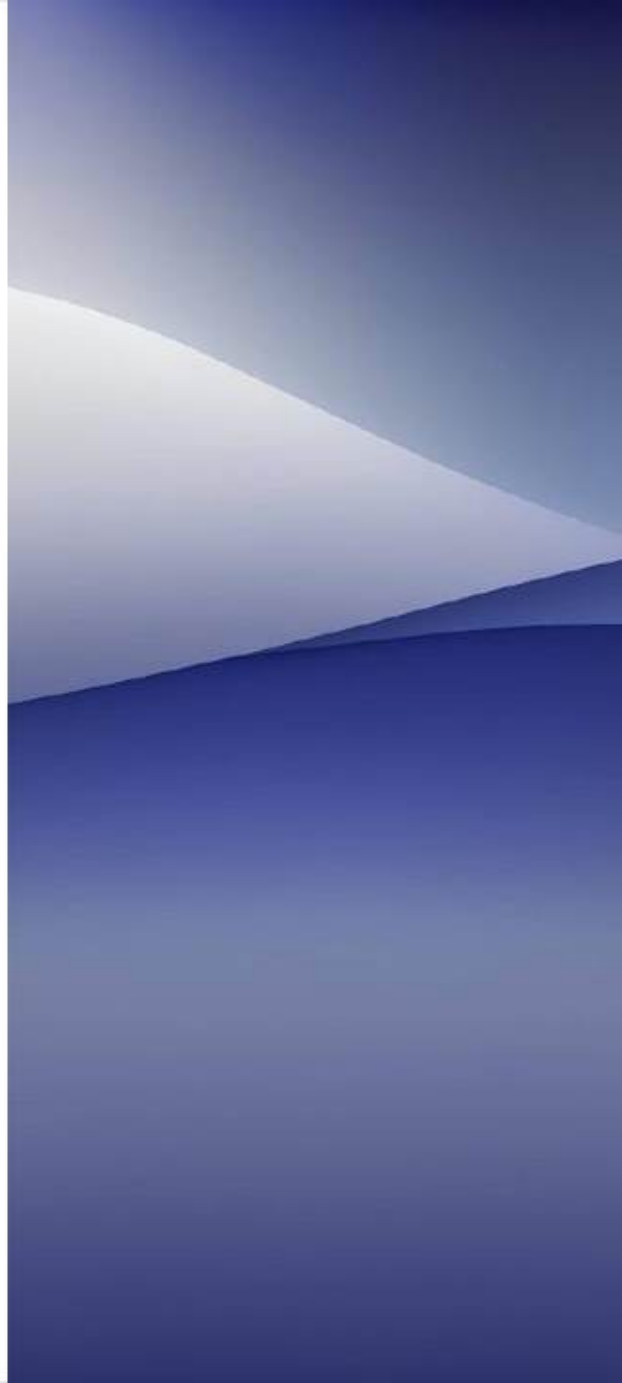
The current system provides generic recommendations based on overall popularity and customer reviews.

2 Missed Opportunities

Customers often miss out on discovering relevant products due to the limited scope of the existing recommendation system.

3 Lack of Personalization

Personal preferences and unique browsing history are not efficiently utilized in the current system.



Proposed System

1

Enhanced Personalization

Our proposed system leverages advanced algorithms to provide highly personalized product recommendations based on individual preferences.

2

Dynamic Updates

Product recommendations are continuously updated in real-time to reflect changing customer preferences and trends.

3

Improved User Experience

Customers will enjoy a seamless browsing experience with tailored recommendations that spark their interest and fulfill their needs.

How It Works

1

Data Collection

Gather customer browsing history, purchase data, and feedback.

2

Pattern Analysis

Identify patterns and correlations between products and customer preferences.

3

Machine Learning

Apply machine learning algorithms to predict personalized recommendations.

4

Real-time Updates

Continuously update recommendations based on new browsing and purchasing behavior.

Advantages

Enhanced Customer Satisfaction

Customers will discover products that truly resonate with their preferences, leading to higher satisfaction levels.

Increased Sales

Personalized recommendations drive impulse purchases and cross-selling opportunities, boosting revenue.

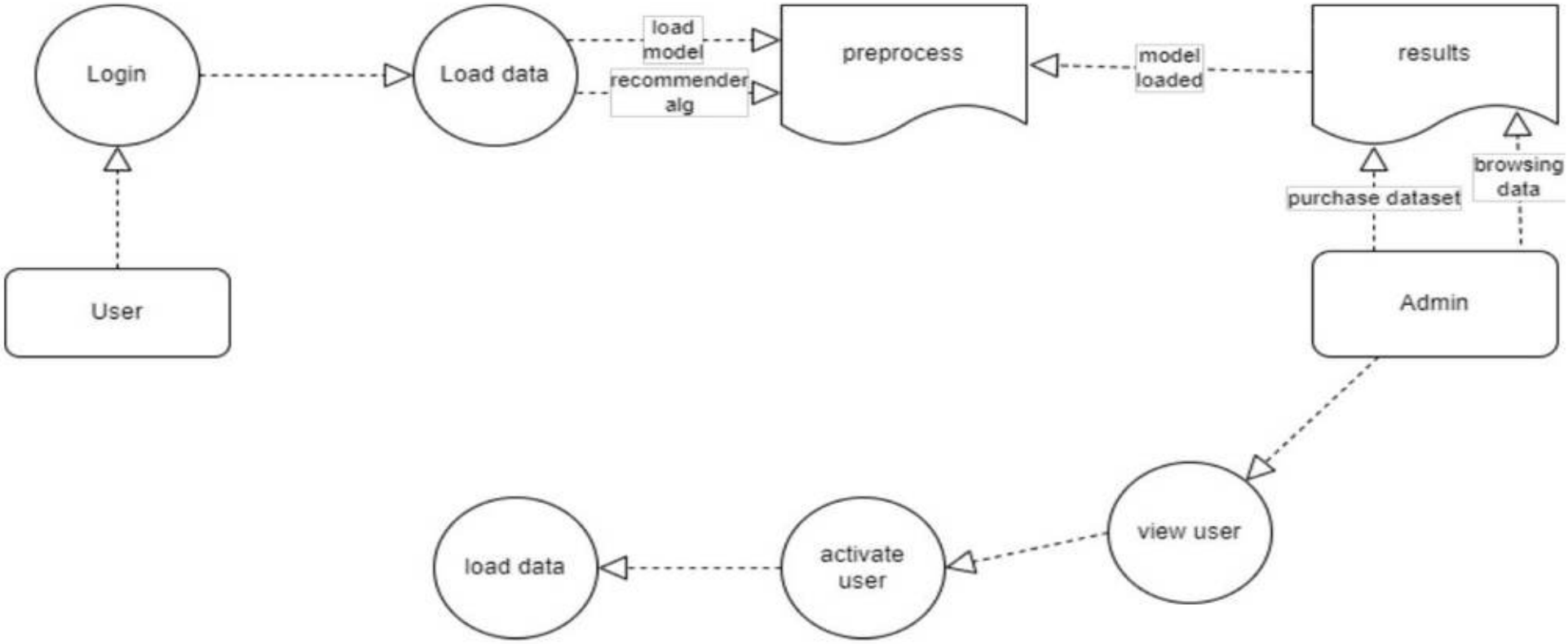
Improved User Engagement

Customers will spend more time on the platform, exploring relevant products and creating a more immersive experience.

Efficient Marketing Spend

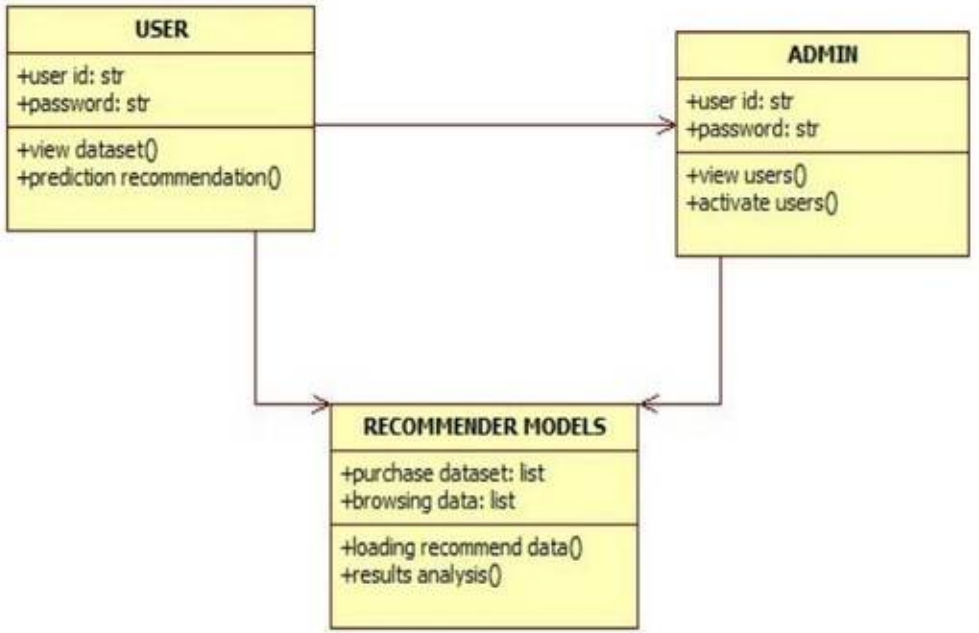
Targeted recommendations optimize marketing efforts by focusing resources on high-potential customers.

Data flow diagram:

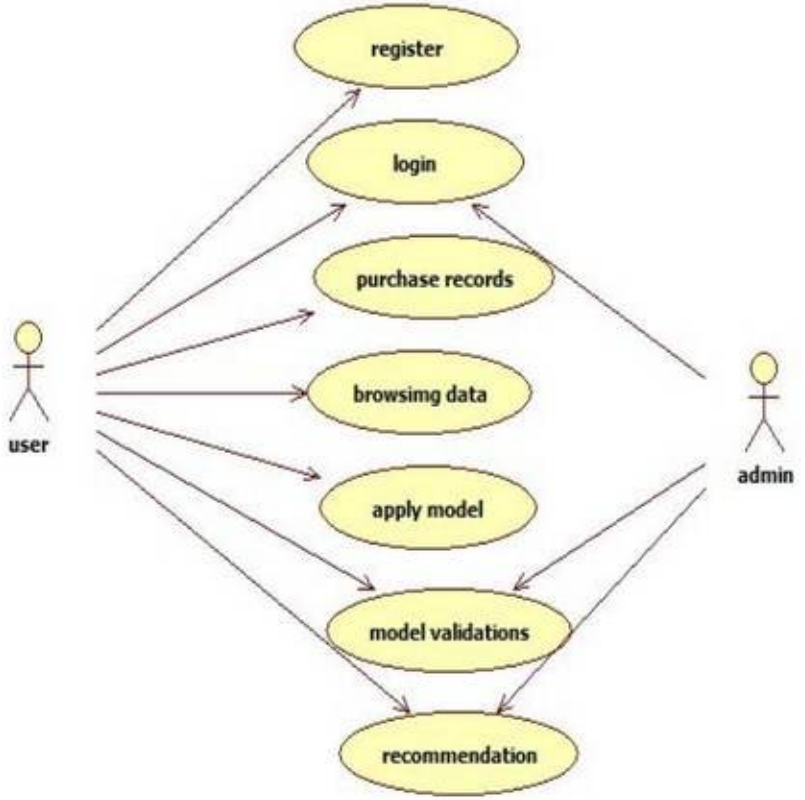


UML Diagrams:

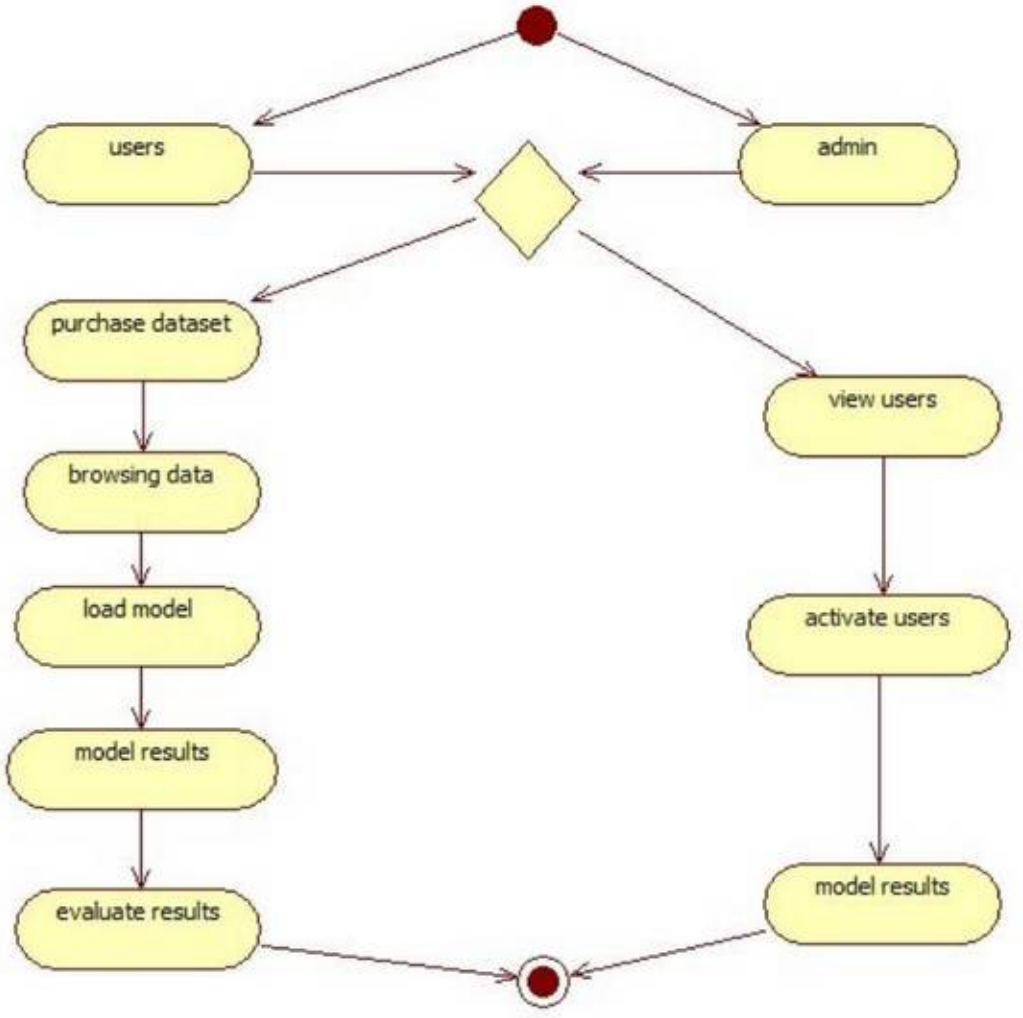
Class diagram:



Use case diagram:



Activity Diagram:



SYSTEM SPECIFICATION:

HARDWARE REQUIREMENTS:

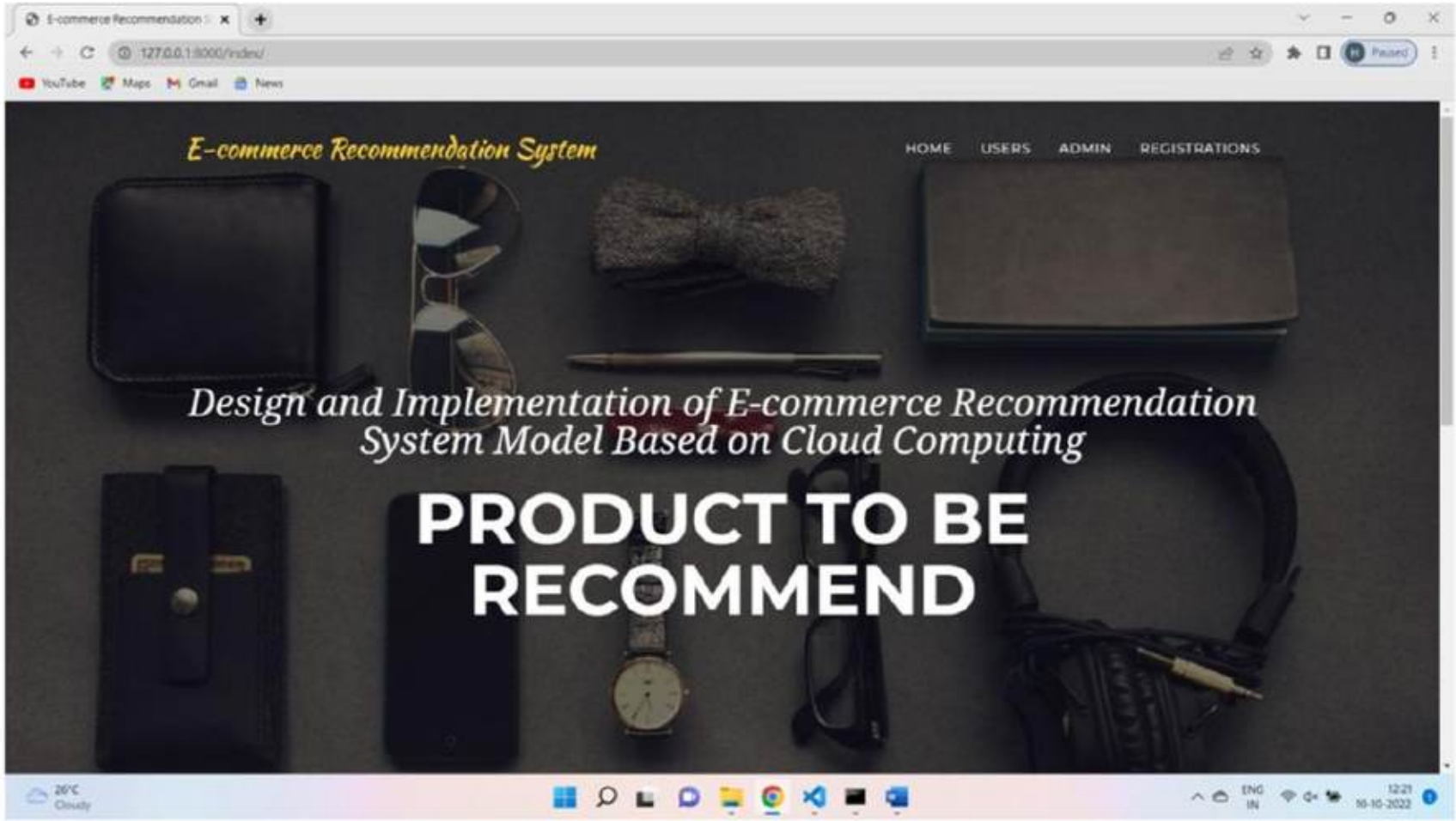
- **System** : Intel i3
- **Hard Disk** : 1TB.
- **Monitor** : 14’ Colour Monitor.
- **Mouse** : Optical Mouse.
- **Ram** : 4GB.

SOFTWARE REQUIREMENTS:

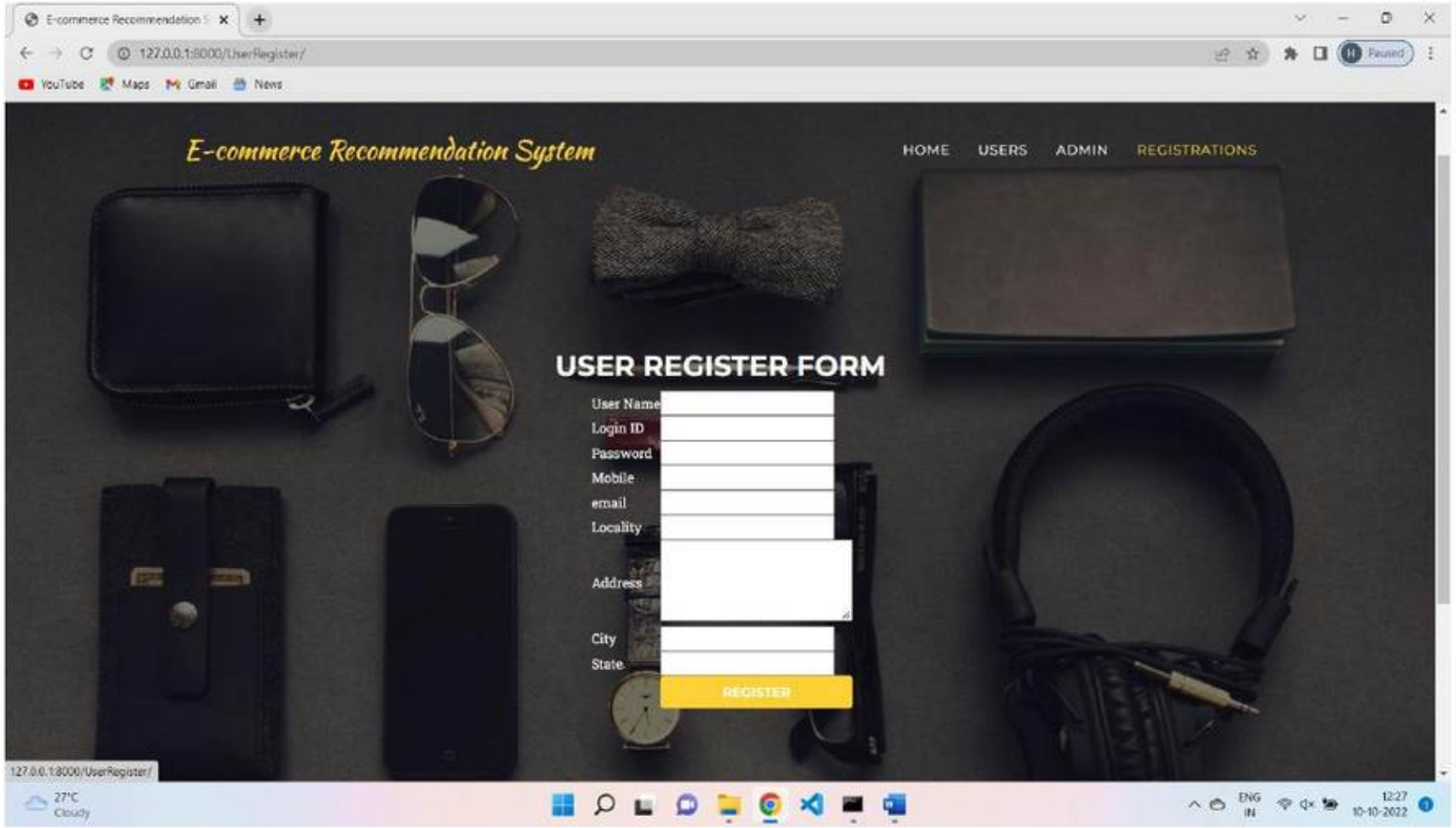
- **Operating system** : Windows 10.
- **Coding Language** : Python.
- **Front-End** : HTML. CSS
- **Designing** : HTML, CSS, JavaScript.
- **Data Base** : SQ Lite.

Screen shots:

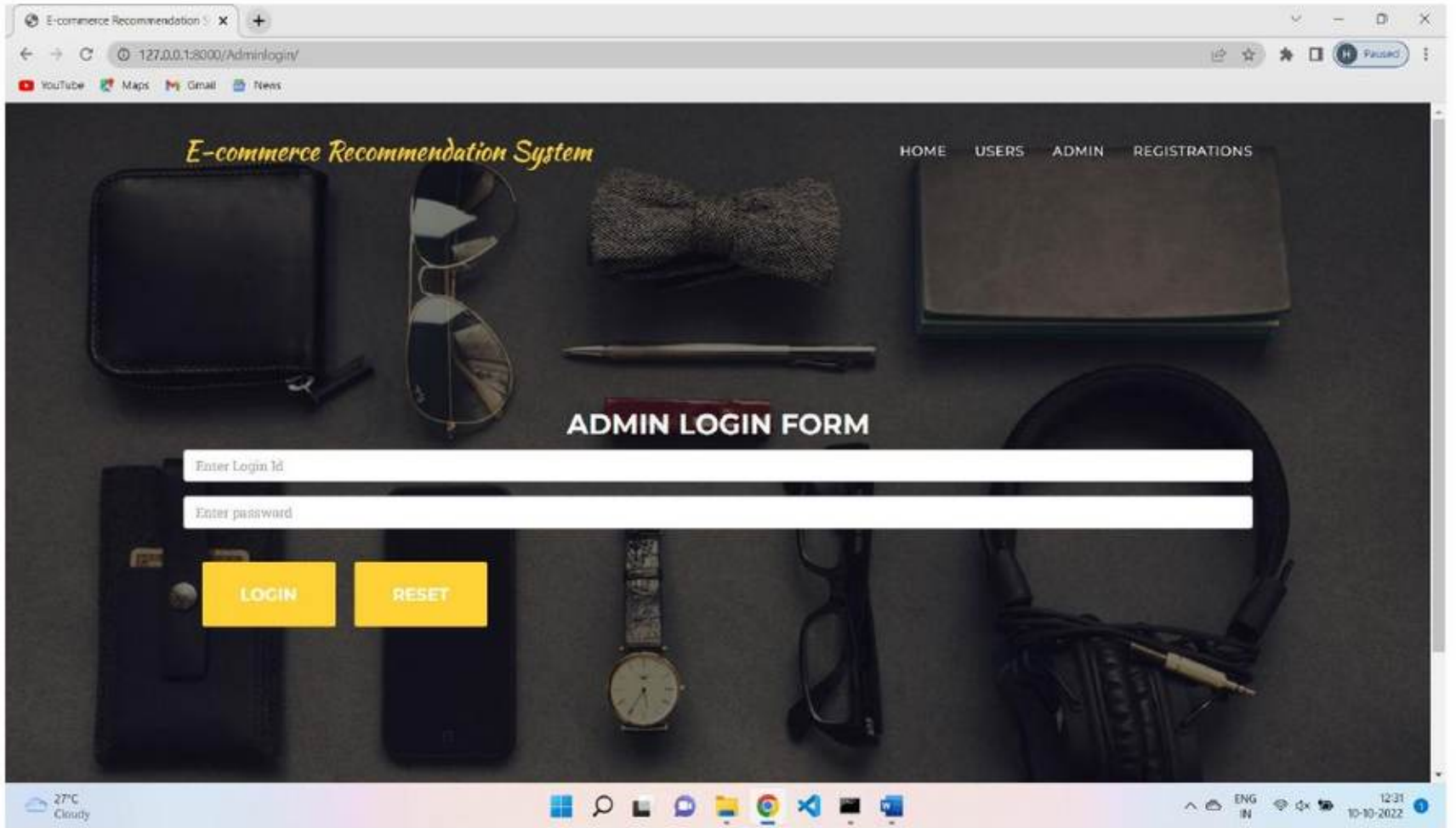
Home page:



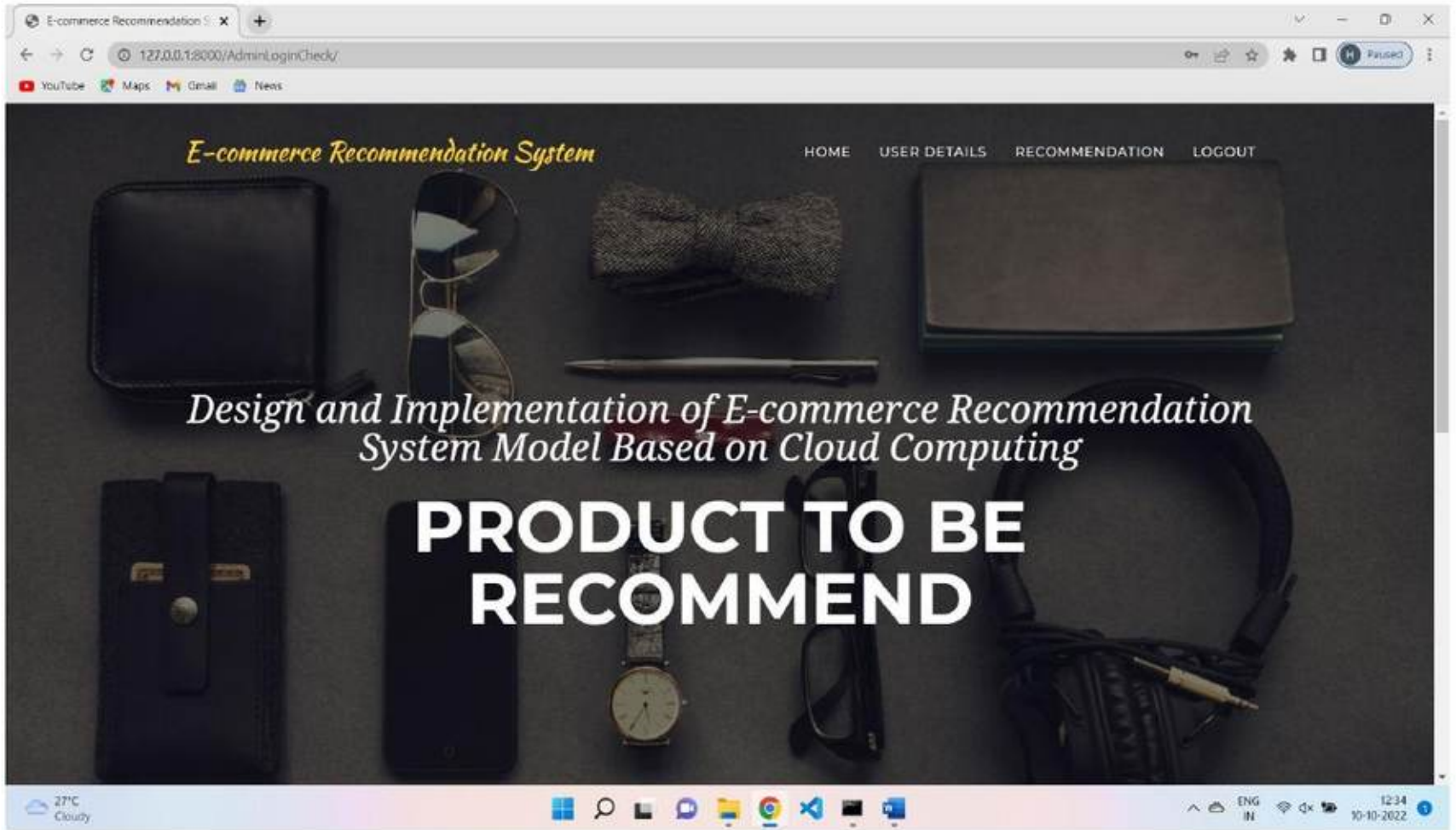
User registration:



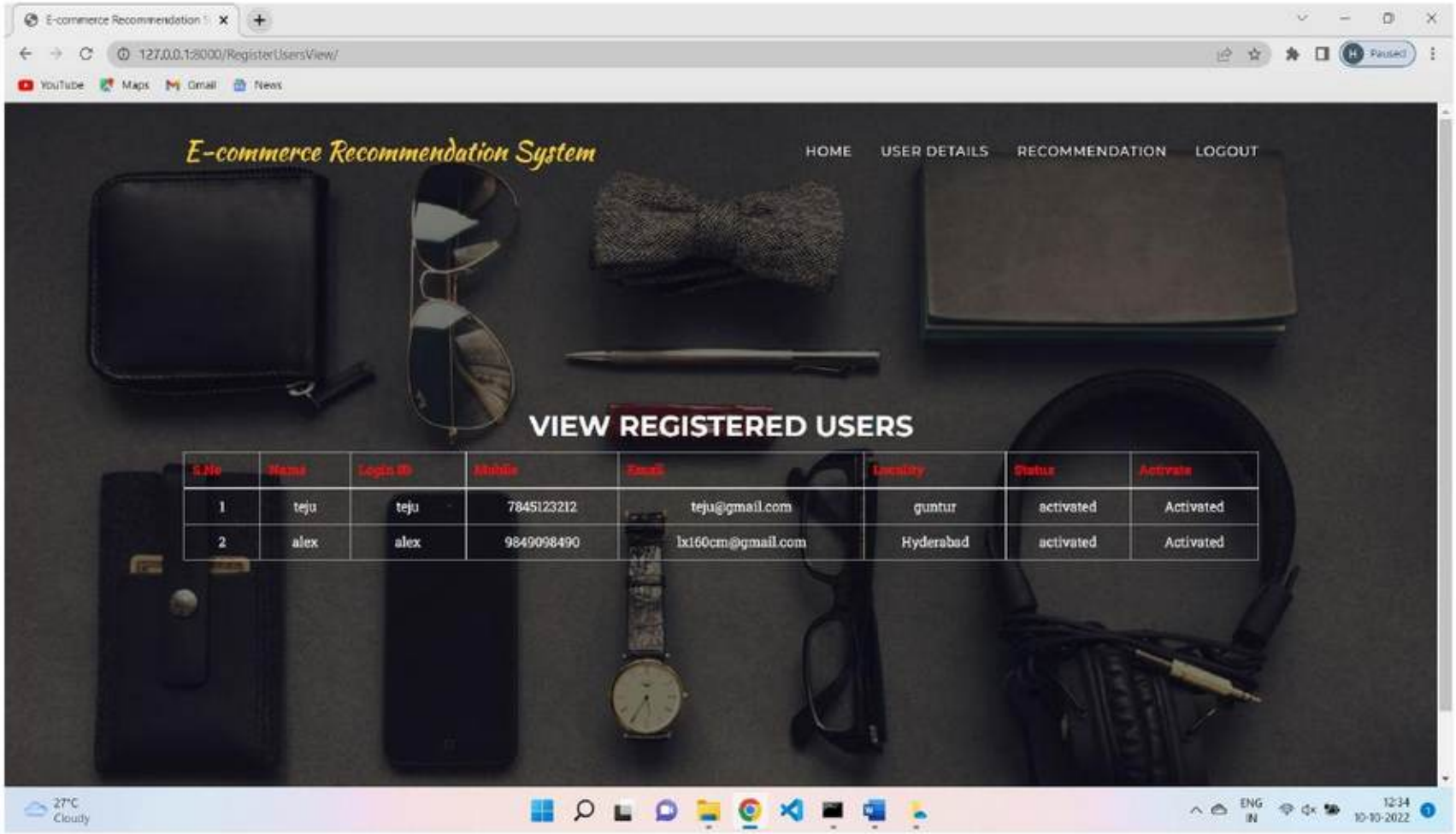
Admin login:


E-commerce Recommendation System
HOME
USERS
ADMIN
REGISTRATIONS
ADMIN LOGIN FORM
Enter Login Id
Enter password
LOGIN
RESET

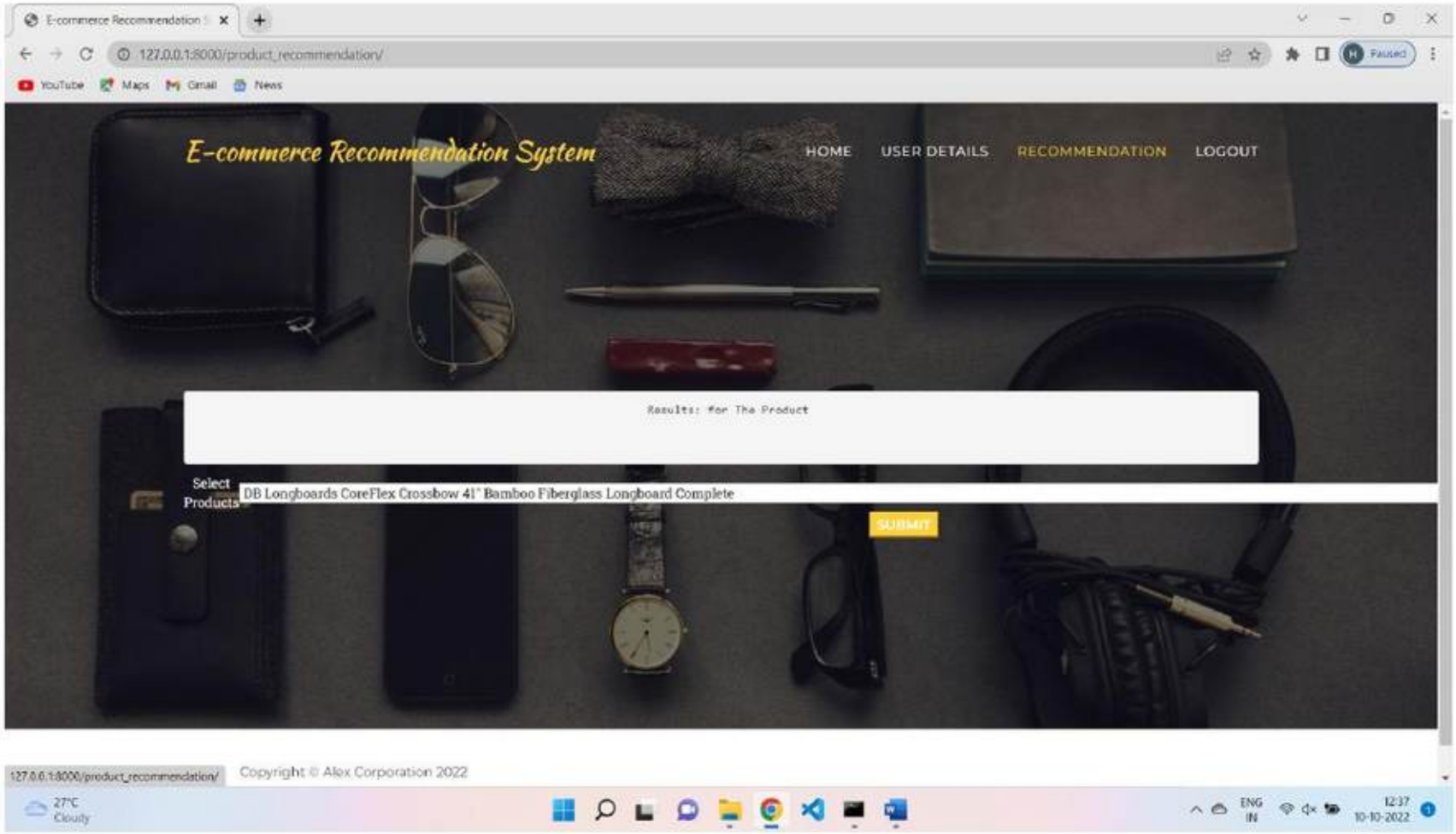
Admin Home:


E-commerce Recommendation System
HOME
USER DETAILS
RECOMMENDATION
LOGOUT
Design and Implementation of E-commerce Recommendation System Model Based on Cloud Computing
PRODUCT TO BE RECOMMEND

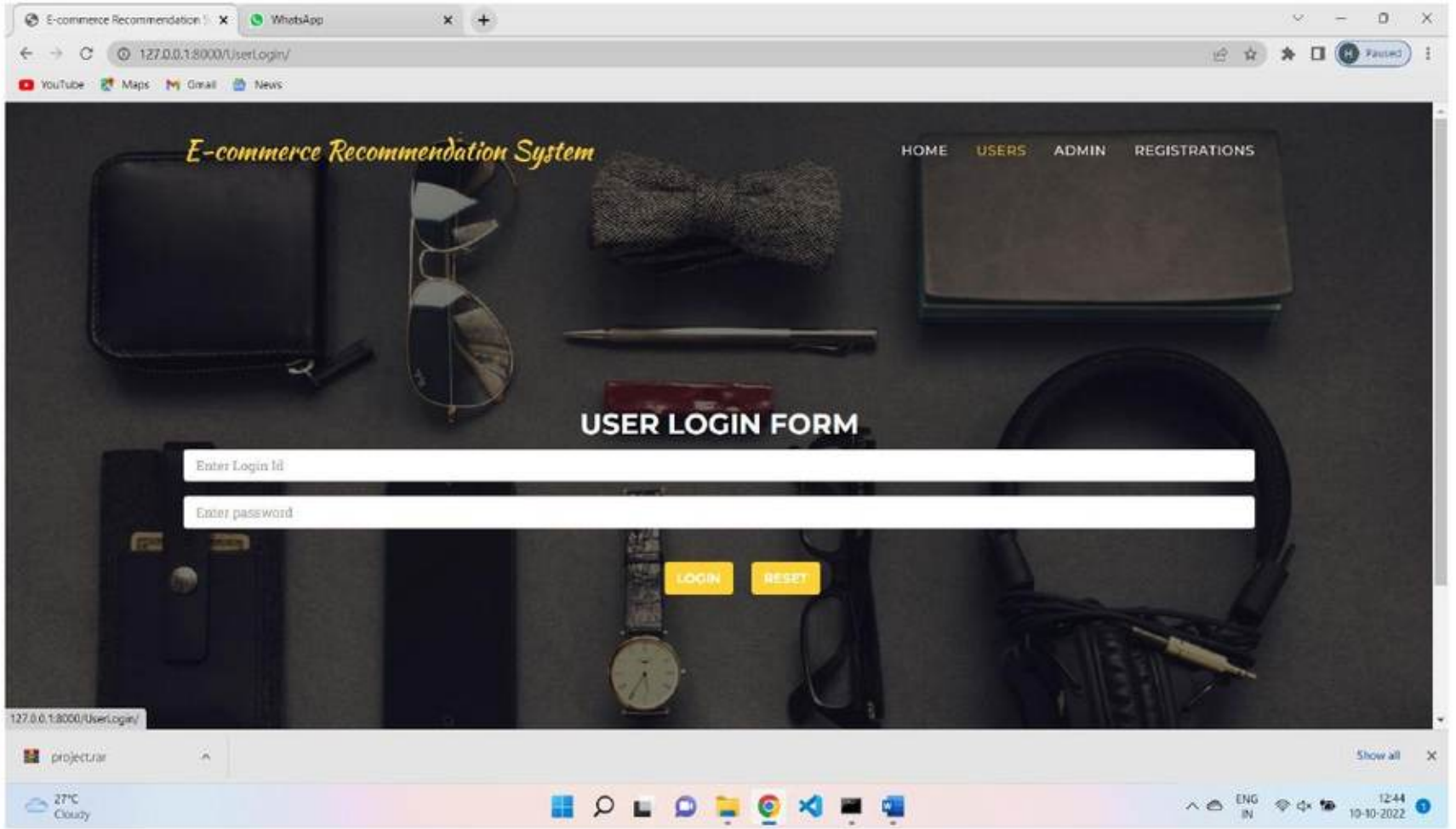
User details



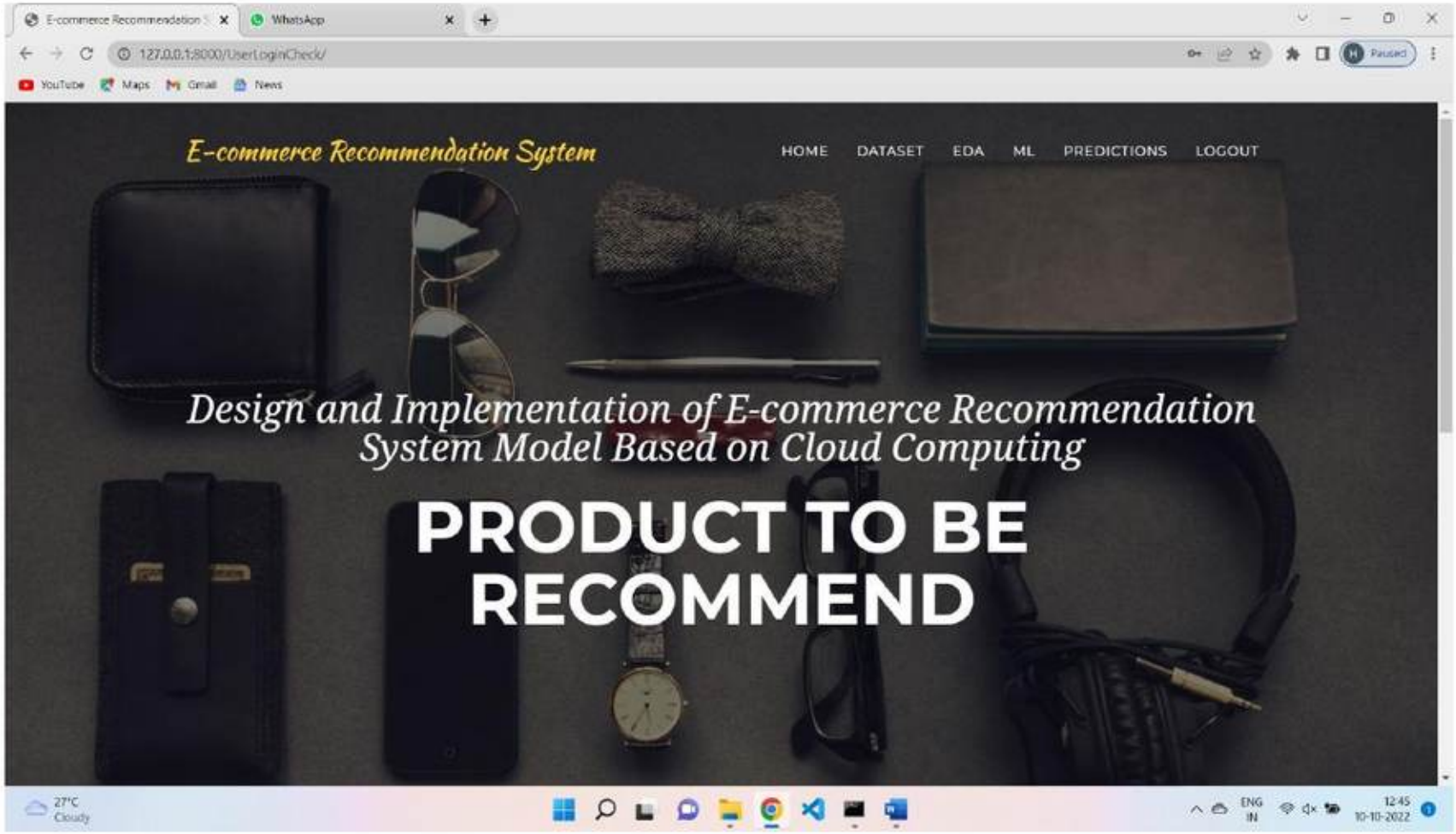
Recommendation:



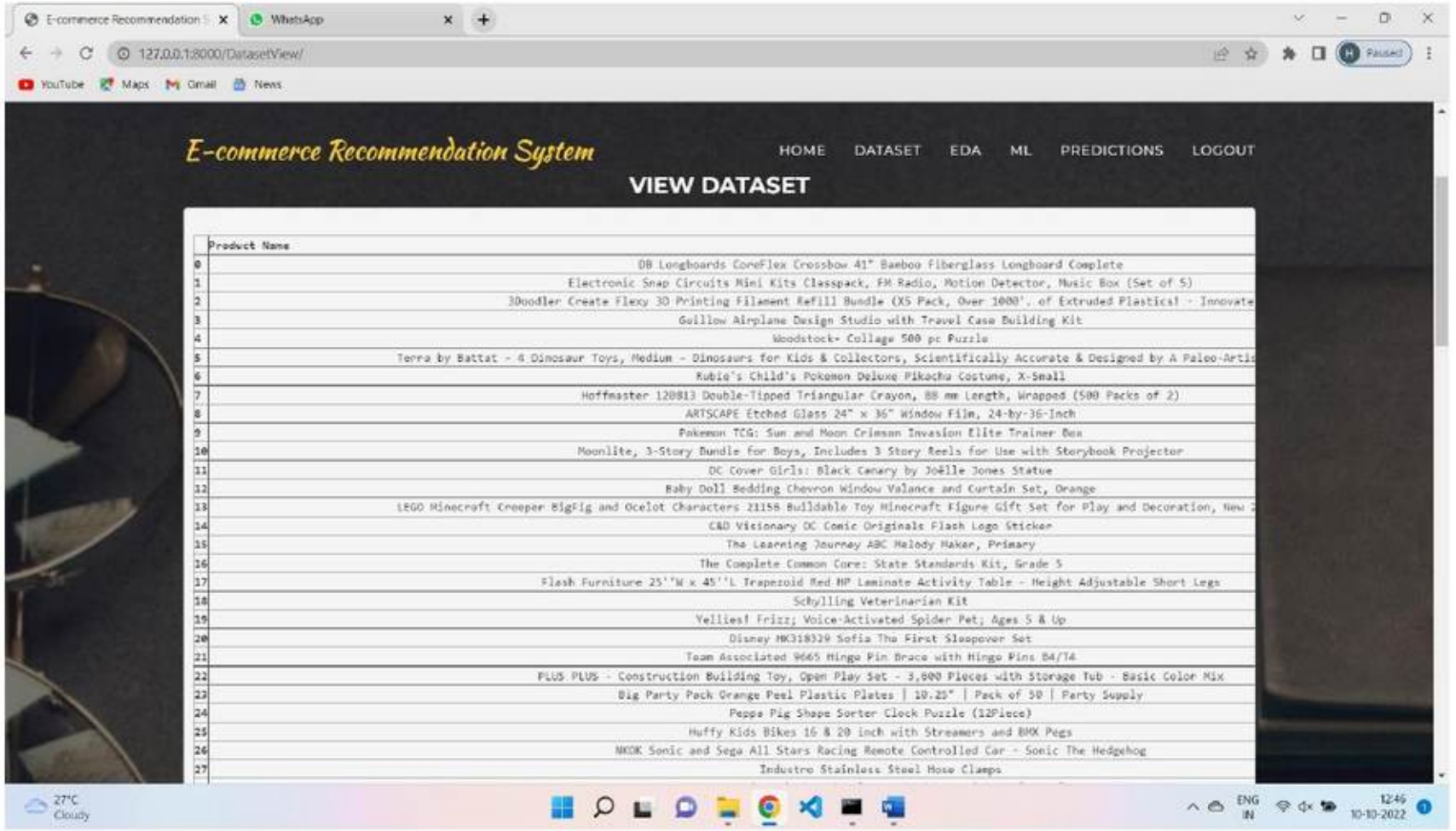
User login:


E-commerce Recommendation System
HOME
USERS
ADMIN
REGISTRATIONS
USER LOGIN FORM
Enter Login Id
Enter password
LOGIN
RESET

User home:

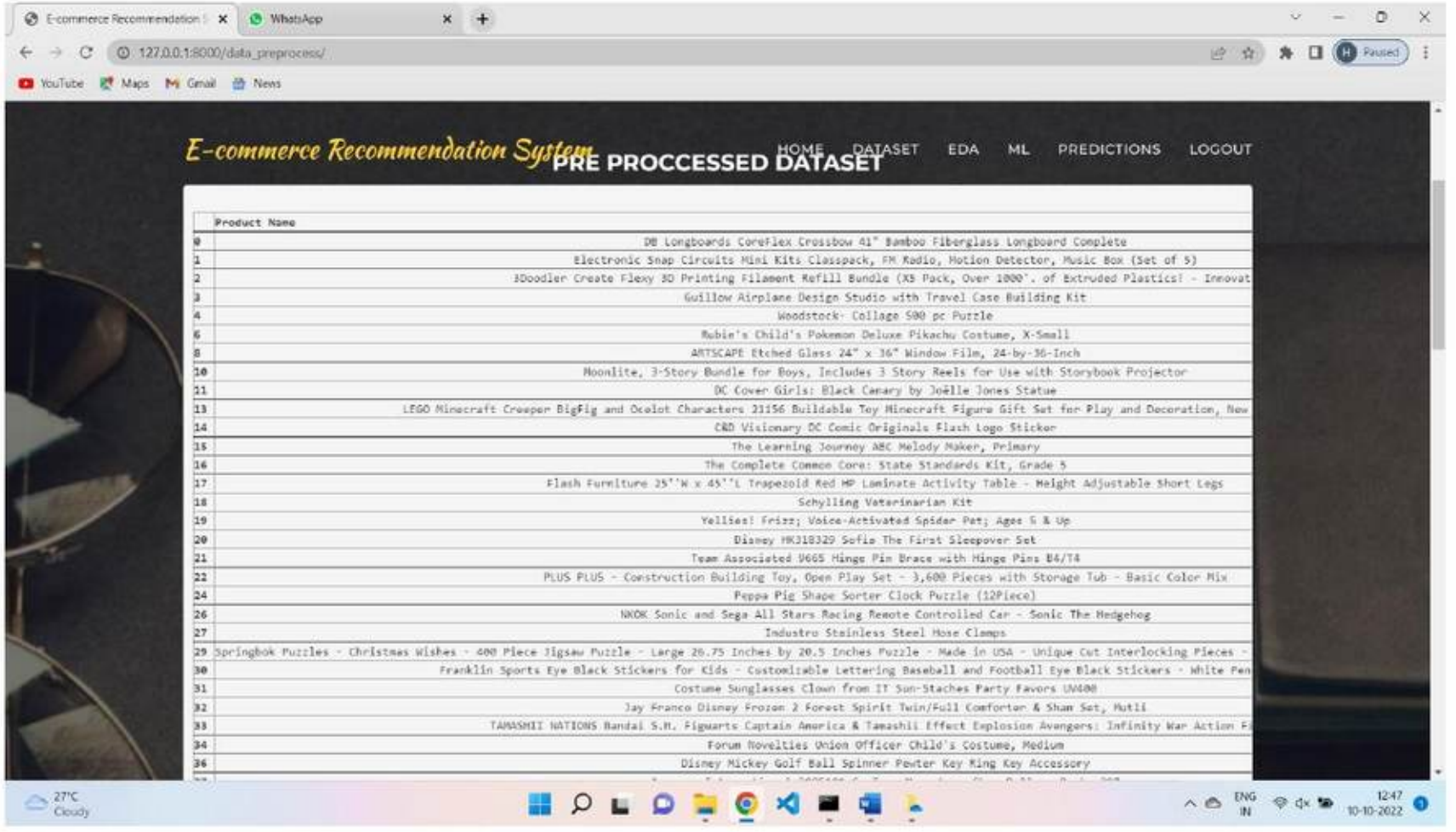

E-commerce Recommendation System
HOME
DATASET
EDA
ML
PREDICTIONS
LOGOUT
Design and Implementation of E-commerce Recommendation System Model Based on Cloud Computing
PRODUCT TO BE RECOMMEND

User dataset:

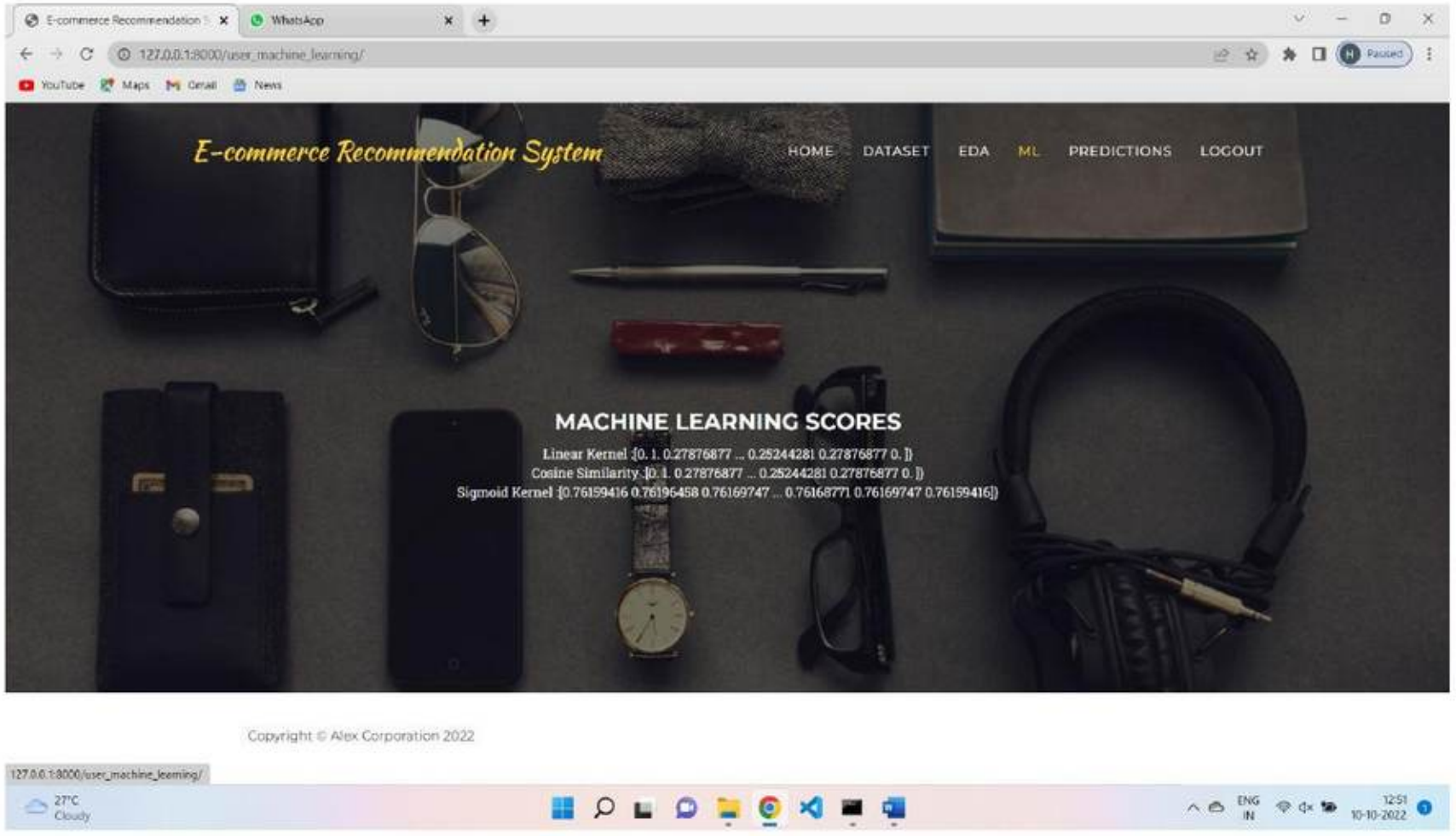


	Product Name
0	DB Longboards CoreFlex Crossbow 41" Bamboo Fiberglass Longboard Complete
1	Electronic Snap Circuits Mini Kits Classpack, FM Radio, Motion Detector, Music Box (Set of 5)
2	3Doodler Create Flexy 3D Printing Filament Refill Bundle (X5 Pack, Over 1000'. of Extruded Plastics! - Innovate
3	Guillow Airplane Design Studio with Travel Case Building Kit
4	Woodstock- Collage 500 pc Puzzle
5	Terra by Battat - 4 Dinosaur Toys, Medium - Dinosaurs for Kids & Collectors, Scientifically Accurate & Designed by A Paleo-Artis
6	Rubie's Child's Pokemon Deluxe Pikachu Costume, X-Small
7	Hoffmaster 120813 Double-Tipped Triangular Crayon, 88 mm Length, Wrapped (500 Packs of 2)
8	ARTSCAPE Etched Glass 24" x 36" Window Film, 24-by-36-Inch
9	Pokemon TCG: Sun and Moon Crimson Invasion Elite Trainer Box
10	Moonlite, 3-Story Bundle for Boys, Includes 3 Story Reels for Use with Storybook Projector
11	DC Cover Girls: Black Canary by Joëlle Jones Statue
12	Baby Doll Bedding Chevron Window Valance and Curtain Set, Orange
13	LEGO Minecraft Creeper BigFig and Ocelot Characters 21156 Buildable Toy Minecraft Figure Gift Set for Play and Decoration, New
14	C&D Visionary DC Comic Originals Flash Logo Sticker
15	The Learning Journey ABC Melody Maker, Primary
16	The Complete Common Core: State Standards Kit, Grade 5
17	Flash Furniture 25''W x 45''L Trapezoid Red HP Laminate Activity Table - Height Adjustable Short Legs
18	Schylling Veterinarian Kit
19	Yellies! Frizz; Voice-Activated Spider Pet; Ages 5 & Up
20	Disney MK318329 Sofia The First Sleepover Set
21	Team Associated 9665 Hinge Pin Brace with Hinge Pins B4/T4
22	PLUS PLUS - Construction Building Toy, Open Play Set - 3,000 Pieces with Storage Tub - Basic Color Mix
23	Big Party Pack Orange Peel Plastic Plates | 10.25" | Pack of 50 | Party Supply
24	Peppa Pig Shape Sorter Clock Puzzle (12Piece)
25	Huffy Kids Bikes 16 & 20 inch with Streamers and BMX Pegs
26	MKDK Sonic and Sega All Stars Racing Remote Controlled Car - Sonic The Hedgehog
27	Industro Stainless Steel Hose Clamps

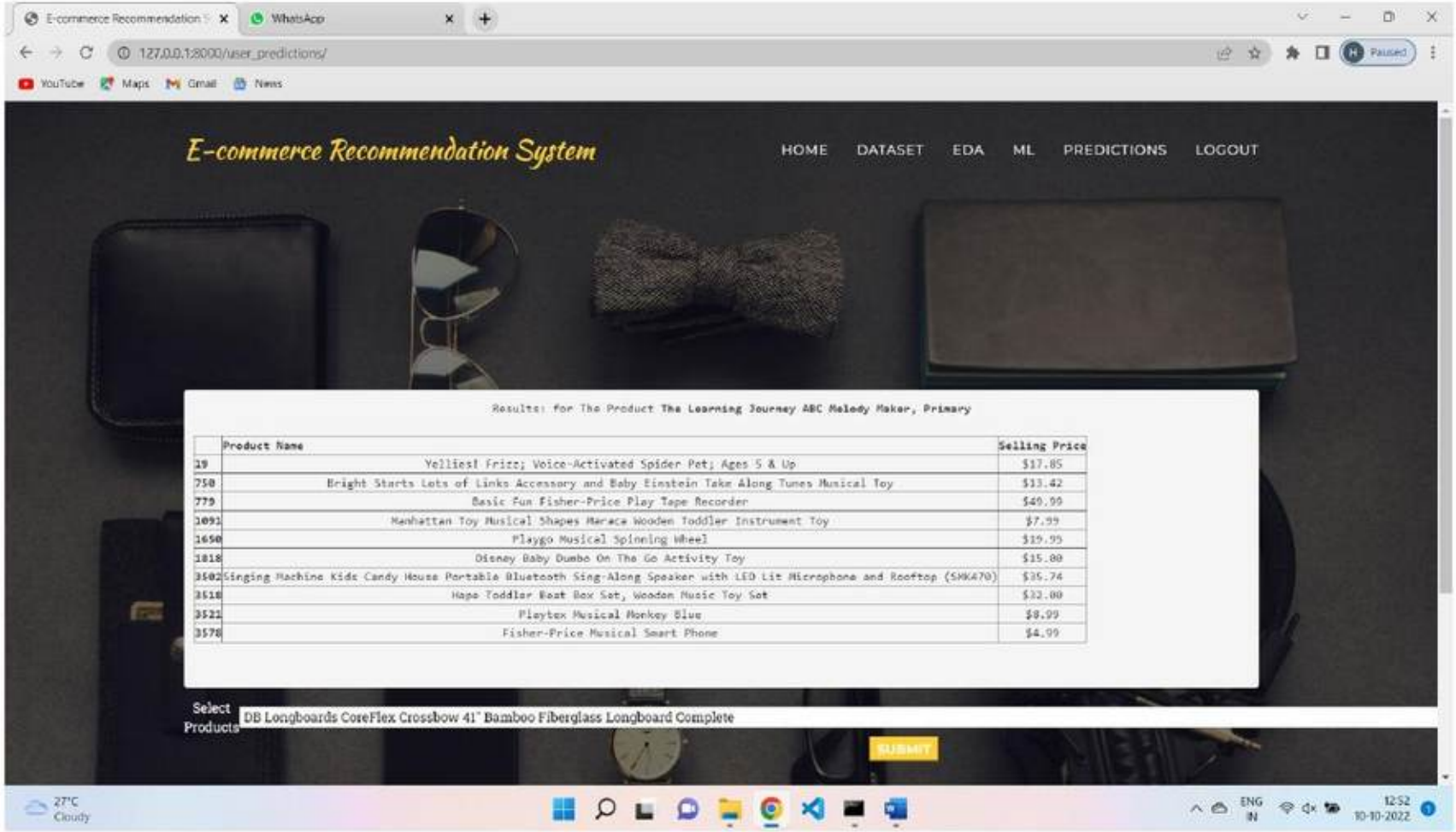
Pre-processed dataset:



ML results:


E-commerce Recommendation System
WhatsApp
127.0.0.1:8000/user_machine_learning/
YouTube
Maps
Gmail
News
Paused
E-commerce Recommendation System
HOME
DATASET
EDA
ML
PREDICTIONS
LOGOUT
MACHINE LEARNING SCORES
Linear Kernel :[0. 1. 0.27876877 ... 0.25244281 0.27876877 0.]
Cosine Similarity :[0. 1. 0.27876877 ... 0.25244281 0.27876877 0.]
Sigmoid Kernel :[0.76159416 0.76196458 0.76169747 ... 0.76168771 0.76169747 0.76159416]
Copyright © Alex Corporation 2022
127.0.0.1:8000/user_machine_learning/
27°C
Cloudy
ENG
IN
12:51
10-10-2022

Prediction of some recommendation result:



Conclusion

E-commerce has become the most mainstream business model in contemporary society. With the explosive growth of information in the era of big data in e-commerce, personalized recommendation system has attracted more and more attention. E-commerce provides more and more shopping and consumption platforms for people. Using recommendation algorithms such as data mining can recommend products that users like to consumers and improve the turnover rate of e-commerce websites. With the rapid development of internet commercial websites, the total amount of information in e-commerce system is increasing day by day, and the problem of information overload is becoming more and more serious. Because the recommendation system is based on user behavior, malicious user behavior should be filtered. Malicious behaviors include users browsing certain products frequently through programs, creating a large number of user behaviors, or businesses adjusting commodity attributes according to algorithms. The powerful storage, operation and security functions of cloud computing, as well as the ideal mode of resource allocation and sharing, have laid a good foundation for the development of e-commerce recommendation engine, resulting in a brand-new business recommendation mode. In the aspect of data integration, the data analysis middleware layer can be used to import the source data of the business system for analysis, and the data of the business system can also be directly stored in the distributed file system layer for management and access.

FUTURE ENHANCEMENT

Using recommendation algorithms such as data mining can recommend products that users like to consumers and improve the turnover rate of e-commerce websites. With the rapid development of internet commercial websites, the total amount of information in e-commerce system is increasing day by day, and the problem of information overload is becoming more and more serious. Because the recommendation system is based on user behavior, malicious user behavior should be filtered

Thank You!

Thank you for joining us in exploring the exciting world of Ecommerce Recommendation Systems. We look forward to revolutionizing your online shopping experience!